



ME 300: Engineering Thermodynamics

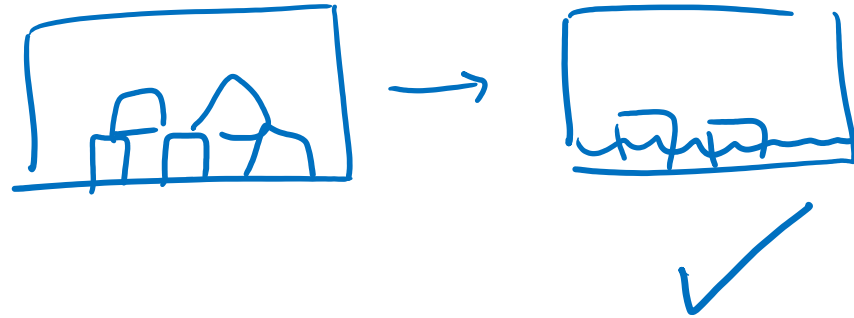
Assumptions about Fluids

Pure substance

Property: quantifiable, macroscopic $\rightarrow$ state = all properties

Pure substance: homogeneous and unchanging chemical composition

$\rightarrow$ allowed: multiple phases of same substance



Simple compressible substance

S.C.S. $\rightarrow$ the following do not change properties

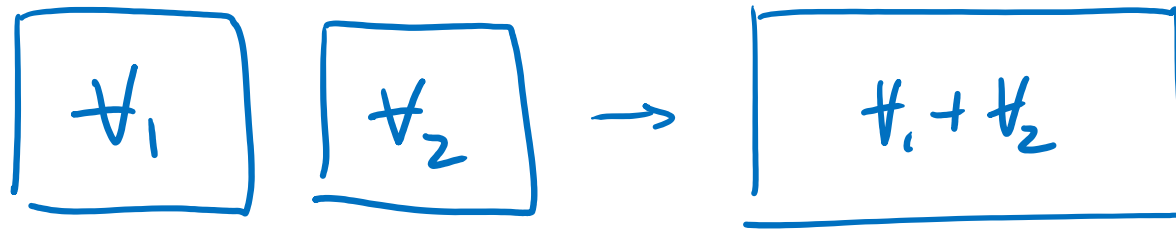
- motion
- fluid shear $\Rightarrow$ bulk
- gravity
- magnetic / electric fields

Extensive properties - M, V, L, N

→ direct function of the extent of system

→ cannot be defined at a point

→ additive

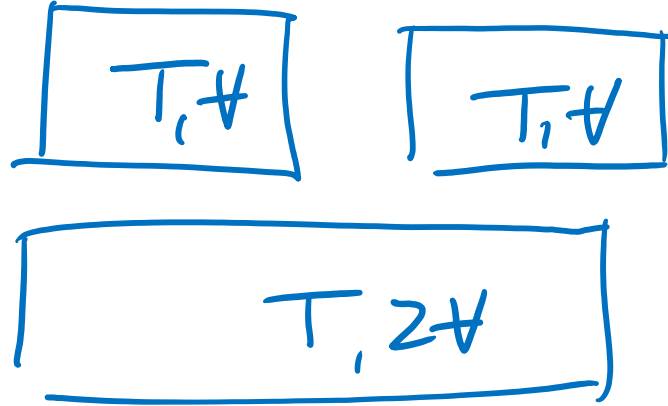


→ ratio of extensive prop = intensive

Intensive properties

→ bulk properties independent of size

- not additive



→ defined at a point

→ $\frac{M}{V} = \rho$